

NetPulse

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Project Title

Website Latency and Packet Loss Monitoring using Streamlit

Objective

The objective of this project is to build a web application that allows users to monitor the latency and packet loss of any given website. The app visualizes the results using a live-updating radial gauge and time-series charts, providing an experience similar to commercial speed test tools.

Tools and Libraries

- Python 3.7+
- Streamlit
- Plotly
- Pythonping
- Pandas
- validators

Features

- User input for website URL
- Continuous pinging at regular intervals
- Display of min, max, average latency (RTT) and packet loss
- Real-time radial gauge visualization
- Time-series plot of latency over time
- Downloadable CSV report of data

Setup Instructions

1. Create a virtual environment:

```
python3 -m venv venv
```

```
source venv/bin/activate # On Unix/Mac
```

2. Install dependencies:

```
pip install --upgrade pip  
pip install -r requirements.txt
```

3. Run the application:

```
streamlit run app.py
```

Conclusion

This project provides a lightweight, user-friendly solution for monitoring website performance. It combines real-time data collection with visualizations, making it useful for both developers and general users.

References

- <https://streamlit.io>
- <https://plotly.com/python>
- <https://github.com/alessandromaggio/pythonping>